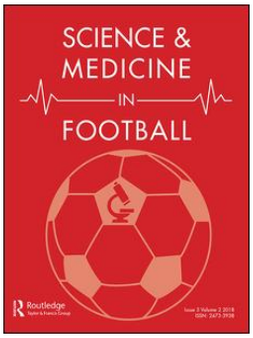

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A systematic review on methodological variation in acute:chronic workload research in elite male football players

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ABSTRACT

Objectives: To investigate and evaluate the methodological variation in research on acute:chronic workload ratio (ACWR) and injury in elite male football players.

Methods: Relevant literature was electronically searched on PubMed, Web of Science, ScienceDirect, and Embase. Additional literature was obtained from studies' bibliographies and authors. Cohort studies investigating the effects of ACWR on male elite footballer injuries were included. Information regarding study population, time frame, protocol, injury classification, and statistical analysis were elucidated.

Results: Database searches led to 2,689 articles. After full text screening, twelve articles remained. All studies were of poor quality. Five studies had GPS-derived workload measures with consideration of running intensity zones, though little consensus over zone thresholds were found. Nine studies incorporated rated perceived exertion data; heterogeneity in exposure type and data collection timing was observed. All studies applied rolling average ACWRs, exploring 1:2, 1:3, and 1:4-week load ratio windows. Differences in data grouping, inference or regression analysis, and other statistical methods were noted.

Conclusion: Existing literature displayed methodological heterogeneity. Future studies should consider consulting guidelines for developing prognostic studies and further examine causal links between workload and injury. From that basis, decisions around ACWR definitions, workload measures, and statistical methods may be more appropriately made.

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KEYWORDS
Soccer; injury prevention; load; methodology; elite performance

Introduction

The evolution of football into a faster and more competitive game reflects an increase in not only the technical, but the physical demands of professional football players (Barnes *et al*. 2014). Nowadays, elite players must endure high match frequencies and periods of congested calendars (Ekstrand *et al*. 2004) raising questions over potential risks involved with meeting these physical demands.

Banister *et al*. (1975) described an athlete's training state as the difference between 'positive' and 'negative' influences (i.e. fitness and fatigue, respectively). With this model, Gabbett *et al*. (2016) recommended the concept of 'acute:chronic workload ratio' to prepare an athlete – if an athlete inadequately trains to develop 'fitness', or workload increases dramatically resulting in 'fatigue', then the athlete is underprepared and susceptible to injury. The term 'workload' as used by Gabbett *et al*. (2016) is commonly applied synonymously with 'load', both referring to the 'dose of training completed by an athlete during an exercise bout'(Wallace *et al*. 2013). Workload involves both internal (physical, physiological, and biomechanical response to exercise; e.g. rated perceived exertion, heart rate or RPE; Impellizzeri *et al*. 2005; Vanrenterghem *et al*. 2017; McLaren *et al*. 2017) and external characteristics (external/environment-related; e.g. distance, velocity, or acceleration; Jaspers *et al*. 2018). By comparing recent workload (e.g. the past week) with workload performed less recently (e.g. the past 2, 3, or 4 weeks) and thereby creating an acute:chronic workload ratio (ACWR), researchers may contextualise the relative spikes in a player's load to investigate the effect of fatigue and fitness on injury risk (Calvert *et al*. 1976; Hulin *et al*. 2014; Gabbett 2016).

Studies have associated harmful effects of injury on team performance, revenue, and success in elite male football (Arnason *et al*. 2004). Lower injury burden and incidence, and higher match availability have all been associated with increased points per league match and with higher final league table ranking in elite male football (Hägglund *et al*. 2013). With an average of two injuries per season per player, professional teams can expect up to 50 injuries a season on a 25-person team, each lasting an average of 37 days (Hägglund *et al*. 2005). In addition, the average cost of a first-team player being injured for 1 month has been estimated to be €500,000 (Ekstrand 2013). As such, teams are finding it increasingly relevant to not only understand injury aetiology, but to produce injury prevention tools. This way, teams aim to organise training that is sufficient to produce positive adaptations but avoids overtraining and injury (Selye 1946; Gabbett and Ullah 2012).

Recently, increasing accessibility of technology, e.g. global positioning software (GPS) has allowed coaches to record players' activity profiles in training and matches (Wehbe *et al*. 2014). This development facilitated routine tracking of training and match workload. As such, several studies made use of the

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available data from training monitoring to examine associations between metrics commonly used for training monitoring and injuries. From these training load measures, new measures of exposure (such as acute workload, chronic workload, and their ratio ACWR) have been created to further investigate the association between workload and injuries.

Although ACWR has been recommended for identifying injury risk (Soligard *et al*. 2016) no guidelines exist for methodological details, including choice of internal and external workload, outcome measures, acute:chronic time windows, and statistical tools. As such, various authors have highlighted outstanding issues present in ACWR and its research, including mathematical coupling (Lolli *et al*. 2017) the inappropriateness of using a ratio (Lolli *et al*. 2018) feasibility of use in the real-world context (Buchheit 2017) and the use of rolling average as opposed to exponentially weighted moving averages (Menaspa 2017). In fact, a recent letter to the Editor of the British Journal of Sports Medicine by Impellizzeri *et al*. (2019) raising several points of contention with a study by Gabbett *et al*. (2016) highlights both the novelty of ACWR as a concept and key uncertainties of its application in football. In addition, the Australian Institute of Sport (Australian Institute of Sport. Athlete Management System. Recent Developments in Athlete Workload and Health Monitoring 2020) has discouraged the use of ACWR to relate activity and injury risk, citing a recent paper by Wang *et al*. (2020) highlighting the limitations of ACWR as a metric for determining injury risk.

Without evaluation of these factors in current football research, conclusions drawn from ACWR research in elite football may be confounded by too large a degree of uncertainty. In addition, previous reviews have tried to summarise results without mentioning or taking into account heterogeneity of methodological approach, which is a crucial aspect to consider prior to results comparison.

This study aims to review the current research methodology on the relationship between ACWR and injury incidence in elite male football. Longitudinal cohort studies that retrospectively analysed prospectively collected data on elite male football players were reviewed, and varying aspects of study methodologies including study population characteristics, time frame, data collection protocol, injury data collection and classification, and statistical methods were evaluated.

Method

The PRISMA statement was consulted prior to the start of this review and the checklist completed (Moher *et al*. 2009). The review methods were established prior to the conduct of the review (including review question, search strategy, inclusion/exclusion criteria and risk of bias assessment) and no significant deviations from the protocol were made.

Search methods for identification of studies

Relevant studies were obtained using computer-based literature search of four databases from inception to 7 July 2019. The databases were as follows: PubMed (1950 – July 2019), Web of Science (1970 – July 2019), ScienceDirect (1995 – July 2019), and Embase (1947 – July 2019). The following search terms were used: *football* OR *soccer* AND *workload* OR *acute:chronic* OR *acute chronic workload* OR *ACWR* OR *load* OR *training intensit** OR *training volume* OR *overtrain** AND *Injur** OR *time loss*. Other relevant studies were identified by three independent reviewers (AW, NH and JH) through searching articles referencing the included studies and the reference lists and author publications from included articles using Google Scholar (Google LLC, Mountain View, California, United States).

Inclusion and exclusion criteria

Original peer-reviewed longitudinal cohort studies in English were included if they examined and reported on: 1) elite male football/soccer players, 2) ACWR (regardless of the time window of either acute or chronic constituent of the ratio) using external and/or internal measures, and 3) the relationship between ACWR and injury rate/risk.

Data collection and analysis

All retrieved studies were downloaded to Endnote Web, V.X4 (Clarivative Analytics, Philadelphia, Pennsylvania, United States). Results were cross-referenced and duplicate studies were deleted. Three reviewers (AW, NH, and JH) independently reviewed titles and abstracts of all identified articles using Rayyan QCRI (Qatar Computing Research Institute, Doha, Qatar) to assess the eligibility of inclusion. A fourth independent reviewer was consulted to settle conflict (KK).

Data extraction

Three reviewers (AW, NH, and JH) independently extracted data from all relevant articles by reading the articles in full. Key areas of interest were elucidated, and the information extracted included:

- Study population (sample size, inclusion/exclusion criteria, loss to follow-up, age range, and league)
- Time frame (year, study duration)
- Study protocol (calculation and definition of ACWR, equipment/tools used, equipment operator, measures of internal/external workload, exposure type measured, and timing and frequency of data collection, handling of loss to follow up/missing data)
- Injury classification (consensus statement used/definition of injury, and how diagnosis/return-to-play was determined)
- Statistical analyses (statistical method applied, data pooling method)

Risk of bias

Studies were assessed using the Newcastle-Ottawa Scale (NOS; Wells *et al*. 2019) by two independent reviewers (AW and KK). The scale was adjusted to maintain relevance to football medicine. For a point to be given for representativeness of intervention and non-intervention cohorts, the population mentioned in the included studies' aims will be compared to that of their participants. For example, a point will be awarded

only if a study with an aim to investigate ACWR and injury risk in elite European football players selected players from several elite European football teams. A point on the NOS was also given for use of self-reported RPE to ascertain exposure. This is due to the quantifiable nature of RPE, its validity and reliability as an indicator of internal load in football (Impellizzeri *et al*. 2004; Haddad *et al*. 2017) and its ubiquity in literature across many sports as a measure of workload (Foster *et al*. 2001; Wallace *et al*. 2009; Manzi *et al*. 2010; Scott *et al*. 2013). A point will also be awarded for the category 'demonstration that interest was not present at start of study' only if studies reported on if there were injured players at the start of the study, and provided a solution for how these data were handled. In the comparability section, a point was only awarded if studies reported characteristics of players with injury compared to those without, such as player position or starting/non-starting status. In assessment of outcome, a point was given only if studies mentioned de-identification of injury data or handover of medical record from healthcare professional to research team. Lastly, the 'adequacy of follow-up' criterion was omitted as only players who left the team were lost to follow-up, a case which is unlikely to introduce outcome bias.

Results

Literature search and study selection

Online search of the four mentioned databases yielded 2,689 results. After removal of duplicates, 2,092 results were screened. Eleven articles remained after title and abstract screening, after which full texts were obtained, eliminating one article as it did not involve ACWR. From the remaining 10 articles, a further two articles were identified from the references lists and included in the review. A final 12 articles were included in this review and read in full for data extraction (Figure 1). These studies included information from different elite leagues, predominantly from Europe, although data from Australia, South America, and Asia are also represented in other studies (Table 2).

Table 1

Quality assessment scores ranged from 2 to 4 out of 7 possible marks. Thus, all included studies were of poor study quality (PQ), marked according to the manual (Wells *et al*. 2019). Only three PQ studies (Delecroix *et al*. 2018, 2019; McCall *et al*. 2018b) explicitly described how injuries at the beginning of their study periods were handled, and only one PQ study (McCall *et al*. 2018b) stated de-identification of injury data.

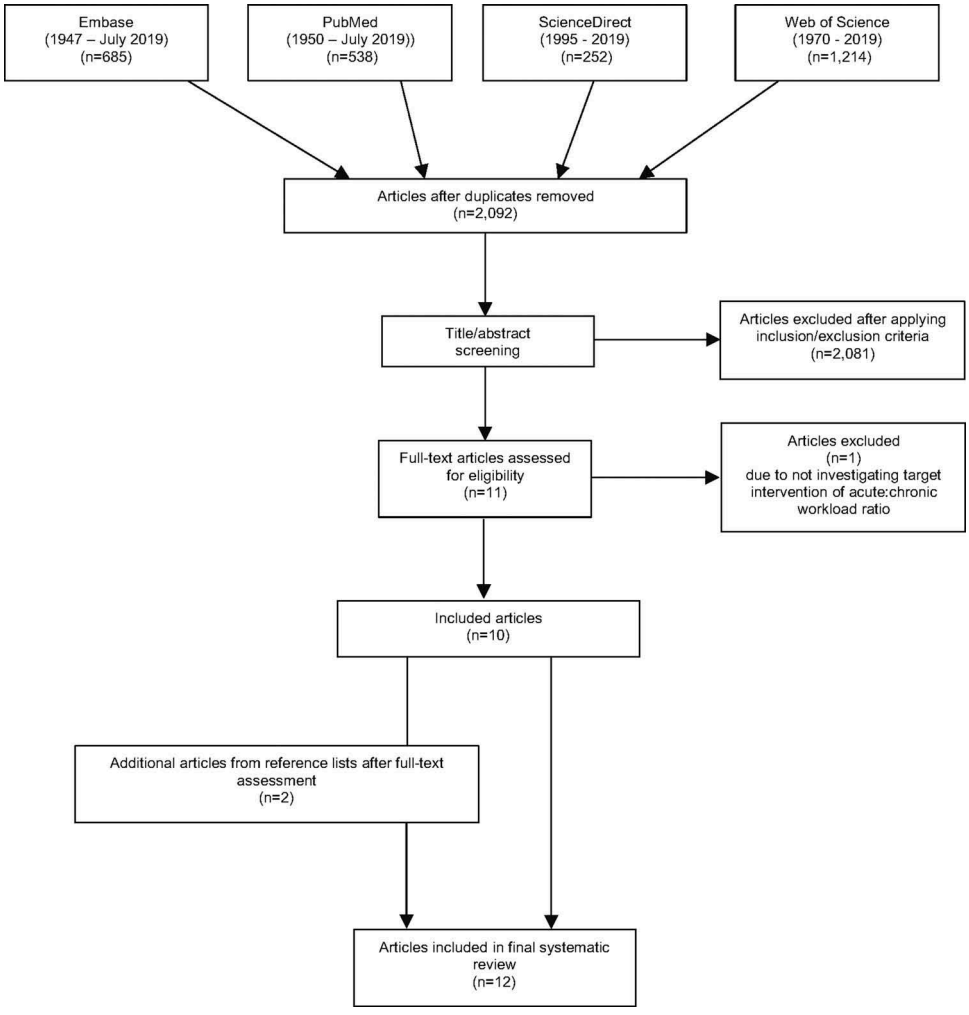

Figure 1. PRISMA flowchart of search strategy.

Table 1. Risk of bias assessment.

Study		Bowen *et al.*[36]	Bowen *et al.*[37]	Delecroix *et al.*[38]	Delecroix *et al.*[39]	Erhmann *et al.*[40]	Fanchini *et al.*[41]	Jaspers *et al.*[42]	Lu *et al.*[43]	Malone *et al.*[44]	McCall *et al.*[45]	McCall *et al.*[46]	Raya-Gonzales *et al.*[47]
Selection (S)	*Representativeness of intervention cohort*	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Selection of the non-intervention cohort											✓	
	Ascertainment of intervention	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Demonstration that interest was not present at start of study			✓	✓						✓		
Comparability (C)	*Comparability of cohorts*												
Outcome (O)	*Assessment of outcome*	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Follow-up duration	2	2	3	3	2	2	2	2	2	4	4	2
Total (S/C/O)		(1/0/1)	(1/0/1)	(2/0/1)	(2/0/1)	(1/0/1)	(1/0/1)	(1/0/1)	(1/0/1)	(1/0/1)	(2/0/2)	(3/0/1)	(1/0/1)
Quality category		POOR	POOR	POOR	POOR	POOR	POOR	POOR	POOR	POOR	POOR	POOR	POOR

None of the included studies were awarded points for comparability.

External workload measures

Six PQ studies (Ehrmann *et al.* 2016; Bowen *et al.* 2017, 2019; Lu *et al.* 2017; Jaspers *et al.* 2018; McCall *et al.* 2018b) measured external workload (Table 3). Five studies (Ehrmann *et al.* 2016; Bowen *et al.* 2017, 2019; Lu *et al.* 2017) used GPS technology; the remaining study (McCall *et al.* 2018b) used number of sessions per week. GPS equipment setup in relevant studies were varied – sampling rates of 5 Hz (Ehrmann *et al.* 2016), 10 Hz (Bowen *et al.* 2017, 2019; Jaspers *et al.* 2018) and 15 Hz (Lu *et al.* 2017) were observed. A 100 Hz triaxial accelerometer sampling rate was applied to 100 Hz in the four PQ studies (Ehrmann *et al.* 2016; Bowen *et al.* 2017, 2019; Lu *et al.* 2017) applying this equipment.

Although all studies using GPS investigated total running distance, heterogeneity was observed in choice and definition of GPS-derived variables (Table 3). Various speed zone groupings included: low intensity (<14.4 km·h^{-1}; Bowen *et al.* 2019), high speed (19.8–25.2 km·h^{-1}; Bowen *et al.* 2019), and sprint (>25.2 km·h^{-1}; Bowen *et al.* 2019); high (>19.7 km·h^{-1}; Ehrmann *et al.* 2016) and very high intensity (14.3–19.7 km·h^{-1}; Ehrmann *et al.* 2016); and low (<14.4 km·h^{-1}; Lu *et al.* 2017), high (>14.5 km·h^{-1}; Lu *et al.* 2017), and very high (>20 km·h^{-1}; Bowen *et al.* 2017; Lu *et al.* 2017; Jaspers *et al.* 2018). Two PQ studies (Bowen *et al.* 2017; Jaspers *et al.* 2018) only investigated distance travelled in one-speed zone (>20 km·h^{-1}). Three PQ studies (Bowen *et al.* 2017, 2019; Jaspers *et al.* 2018) quantified acceleration efforts, with one of these studies (Jaspers *et al.* 2018) defining acceleration as movement >1 m·s^{-2}, whereas two studies (Bowen *et al.* 2017, 2019) defined it as *'a change in speed for >0.5 seconds with maximum acceleration in that period >0.5 m·s^{-2}'*. Further, two PQ studies (Jaspers *et al.* 2018; Bowen *et al.* 2019) quantified deceleration efforts. Additional variables included mean speed (Lu *et al.* 2017), body load/new total body load (Ehrmann *et al.* 2016; Bowen *et al.* 2017; Lu *et al.* 2017), meters per minute (Ehrmann *et al.* 2016), and exposure (min; McCall *et al.* 2018a). All studies collected data from every session, though two PQ studies (Lu *et al.* 2017; Jaspers *et al.* 2018) could not collect data from FIFA games due to the federation's restrictions on GPS use.

Internal workload measures

Nine PQ studies (Lu *et al.* 2017; Malone *et al.* 2017; Delecroix *et al.* 2018, 2019; Fanchini *et al.* 2018; Jaspers *et al.* 2018; Raya-González *et al.* 2018; McCall *et al.* 2018a, 2018b) measured internal workload, with all nine studies using the sRPE method (multiplying session duration by RPE score (Borg *et al.* 1987); Table 3). Eight of these studies based their RPE scales on a modified Borg scale (Borg *et al.* 1987) with the remaining study (Raya-González *et al.* 2018) using Foster's scale (Foster *et al.* 2001). Differences between studies were most prominent in exposure types measured. Only three PQ studies (Malone *et al.* 2017; Delecroix *et al.* 2018; McCall *et al.* 2018a) specified

Table 2. Study and Player Characteristics.

Study	Study period	Duration (seasons)	Matches observed	Training sessions observed	N	Age (y)	Mass (kg)	Height (cm)	Level/League	Inclusion criteria	Exclusion criteria	Participants lost/gained
Bowen *et al.*[36]	2013–15	2	12,117 total		32	17.3 ± 0.9	74.1 ± 7.0	180.0 ± 7.3	Elite youth (U18/U21), EPL Category 1 Academy	Full-time	Goalkeepers	-
Bowen *et al.*[37]	2014–17	3	10,221 total		33	25.4 ± 3.1	79.9 ± 7.7	182.0 ± 6.9	EPL, England	Full-time	Goalkeepers	-
Delecroix *et al.* [38]	-	1	2 games per w	-	130	-	-	-	UECL, UEL level (French Ligue 1, Spanish La Liga, Italian Serie A, EPL)	First team	Players leaving club (excluded from time of leaving) Pre-existing injury at start of study	18 lost 21 gained
Delecroix *et al.*[39]	-	U19: 4 U21: 5	-	-	122	U19: 16.8 ± 0.9 U21: 20.1 ± 0.3	70.9 ± 7.3	178.6 ± 6.8	Elite, French Ligue 1	All players (full-time)	Players leaving club (excluded from time of leaving) Pre-existing injury at start of study	-
Erhmann *et al.*[40]	18 July 2011–30 March 2012	1 (incl. pre-season)	28	-	19	25.7 ± 5.1 (18–33)	77.3 ± 4.2	181.0 ± 4.8	Professional elite, Hyundai A-League, Australia	Full-time	Injured players Players on national team duty Injuries that cannot be pinpointed to a game or training session Competitive matches	No player loss. 3 pre-game training sessions excluded due to logistical difficulties.
Fanchini *et al.*[41]	2013–2016	3 (incl. pre-seasons)	2013–14: 494 h 2014–15: 592 h 2015–16: 593 h	2013–14: 4146 h 2014–15: 4309 h 2015–16: 4134 h	34	26 ± 5	78 ± 4	182 ± 5	Professional, Serie A, Italy	Players who competed in at least 1 season	-	-
Jaspers *et al.*[42]	2014–2016	8102 (incl. pre-seasons)	2 matches and on-field training sessions		35	23.2 ± 3.7	77.5 ± 7.4	182 ± 6	Professional, Eredivisie League, Netherlands	First team	Goalkeepers Weeks when players were called to national team duty Off-season periods	-
Lu *et al.*[43]	2013–2015	3 (incl. pre-seasons)	211 ± 55 sessions per player		45	26.4 ± 5.1	74.5 ± 12.1	181.3 ± 7.1	Professional, Hyundai A-League, Australia, & ACL	-	Goalkeepers	-
Malone *et al.*[44]	-	1 (incl. pre-season)	-	460 sessions from 18 July 2011–30 March 2012	48	25.3 ± 3.1	72 ± 7	183 ± 7	Professional, Highest level of European competition	-	-	-
McCall *et al.*[45]	2014–2015	4 w prior to and during training camps of international tournaments	-	-	33	Brazil: 26.6 ± 4.7 Asian: 26.1 ± 3.5	Brazil: 77.5 ± 7.2 Asian: 77.3 ± 6.3	Brazil: 182.4 ± 7.2 Asian: 183.3 ± 7.0	National team, 2014 Brazil WC & 2015 AC	-	Goalkeepers Players with inconsistent data	2 outfield players did not make squad in Brazil. 2 outfield players left due to injury. 3 players incurred contact injury in first week
McCall *et al.*[46]	2015–2016	1 season (incl. pre-season)	-	-	171	25.1 ± 4.9	77.5 ± 7.2	181.6 ± 6.7	Elite, UECL	-	Untrustworthy/inconsistent data reported by main contact staff	1 team excluded due to inconsistent data. 29 players further excluded.
Raya-Gonzales *et al.*[47]	2016–2017	1	38 (1 per w)	141 (4 per w)	22	18.6 ± 0.6	72.2 ± 6.9	178 ± 4	First division U19, La Liga, Spain	-	Players not part of the team for at least 25 weeks	-

EPL, English Premier League; UECL, UEFA Champions League; UEL, UEFA Europa League; ACL, Asian Champions League; WC, World Cup; AC; Asian Cup

Table 3. Methodological details.

Study	Exposure measured: *On-pitch training*	*Gym/ Individual-based*	*Recovery/ Rehab*	*Competitive match*	*Friendly/ Pre-season*	Int. WL measure	Participant familiarisation	Ext. WL measure	Equipment used	Equipment operator	Frequency of data collection	Timing of data collection
Bowen *et al.*[36]	✓	-	-	✓*		-	-	• TD • HSD (>20 km·h^{-1}) • TL • ACC	• 10 Hz GPS + 100 Hz accelerometer (Viper V.2, StatsSports, Ireland) • Specialised analysis software (Viper, V.2.1.3.0)	-	All sessions	Following each session
Bowen *et al.*[37]	✓	-	-	✓$^{\Psi}$	✓	-	-	• TD • LID (<14.4 km·h^{-1}) • HSD (19.8–25.2 km·h^{-1}) • SD (>25.2 km·h^{-1}) • ACC • DEC	• 10 Hz GPS + 100 Hz accelerometer (Viper 2, StatsSports, Ireland) • Specialised analysis software (Viper, V.2.1.3.0) • Semi-automated camera system video/VID (TRACAB; ChryonHego, New York, USA)**	-	All sessions	Following each session
Delecroix *et al.*[38]	✓	✓	✓	✓	-	sRPE = RPE (mod. Borg CR10)[46] x duration[30]	-	-	-	Same science or medicine department staff member	All sessions	30 min after session
Delecroix *et al.*[39]	✓	-	-	✓	-	sRPE = RPE (mod. Borg CR10)[46] x duration	-	-	-	Sport clinician	All sessions	30 min after session
Erhmann *et al.*[40]	✓	✓	✓	-	✓	-	-	• TD • HIR (14.3–19.7 km·h^{-1}) • VHIR (>19.7 km·h^{-1}) • MpM • New total body load	• 5 Hz GPS + 100 Hz triaxial accelerometer (SPI-Pro GPSports, Canberra, Australia) • GPSports Team AMS Software (GPSports)	-	All sessions	Immediately after session
Fanchini *et al.*[41]	✓		✓*	✓	✓	sRPE (mod. Borg CR10)	-	-	-	-	All sessions	Within 30 min after session
Jaspers *et al.*[42]	✓	✓	-	✓	✓	sRPE = RPE (mod. Borg CR10) x duration	✓	• TD • THSR (>20 km·h^{-1}) • ACCeff (>1 km·s^{-2}) • DECeff (<-1 km·s^{-2})	• 10 Hz GPS (MinimaxS4 and Optimeye S5, Catapult Sports, Melbourne, Australia)	-	Int. WL = all sessions, Ext. WL = all sessions excl. 2014–15 FIFA matches	Int. WL = 30 min after training
Lu *et al.*[43]	✓		✓*	✓ *		sRPE (Borg CR10) x duration[28]	-	• TD • Distance by speed zones (low: <14.4 km·h^{-1}; high: >14.5 km·h^{-1}; very high: >20 km·h^{-1}) • Mean speed • Body load • Exposure (min)	• 15 Hz GPS + 100 Hz, 16 G triaxial accelerometer (SPI-HPU GPSports, Canberra, Australia)	Sport science and conditioning staff	Int. WL = all sessions Ext. WL = all sessions excl. gym or individual-based sessions	Int. WL = 30 min after session
Malone *et al.*[44]	✓	✓	✓	✓	✓	sRPE = RPE (mod. Borg CR10) x duration	-	-	• Custom designed application on a portable computer table (iPad, Apple Inc, California, USA)	-	All sessions	Immediately after session
McCall *et al.*[45]	✓	✓	✓	✓	-	sRPE = RPEx duration	✓	Number of sessions per w	• Smartphone/tablet application (Kinetic Athlete, Canberra, Australia)	Fitness coach	All sessions	Within 30 min after session

(*Continued*)

Table 3. (Continued).

Study	Exposure measured					Int. WL measure	Participant familiarisation	Ext. WL measure	Equipment used	Equipment operator	Frequency of data collection	Timing of data collection
	On-pitch training	*Gym/ Individual-based*	*Recovery/ Rehab*	*Competitive match*	*Friendly/ Pre-season*							
McCall *et al.*[46]	✓	✓	✓	✓	✓	sRPE = RPE (Borg CR10) x duration	-	-	-	Instruction sent to teams to standardise data collection. Same science/ medicine staff member	All sessions	30 min after session
Raya-Gonzales *et al.*[47]	✓	✓*		✓	✓	sRPE (Foster's scale)	✓	-	-	Same physical trainer	All sessions	30 min after session

Ext. = external, Int. = internal; Mod. = modified; WL = workload; Ψ = only from 2015 onwards; * = unspecified nature of exposure type (competitive/friendly match; individual-based gym/recovery session); ϒ = except GPS data for FIFA matches due to restrictions on use of GPS units; TD = total distance; HSD = high speed running; TL = total load; ACC = acceleration efforts; LID = distance in low intensity running; HSD = distance in high speed running; SD = sprinting distance; DEC = deceleration efforts; HIR = high intensity running; VHIR = very high intensity running; MpM = meters per minutes; THSR = total high speed running; ACCeff = acceleration efforts; DECeff = deceleration efforts.

collecting data from all exposure types (on-pitch training, gym, recovery, friendly matches, and competitive matches). Three PQ studies (Lu *et al*. 2017; Jaspers *et al*. 2018; Delecroix *et al*. 2019) did not collect data from recovery/rehabilitation sessions, and one PQ study (Delecroix *et al*. 2019) did not collect data from gym sessions. Three PQ studies (Lu *et al*. 2017; Fanchini *et al*. 2018; Raya-González *et al*. 2018) did not specify whether training data included gym and recovery sessions, and four studies (Bowen *et al*. 2017; Lu *et al*. 2017; McCall *et al*. 2018b; Delecroix *et al*. 2019) did not specify whether data collected from matches were friendly or competitive matches. All studies collected data from every session of the exposure type mentioned in their respective data collection protocols. Six PQ studies (Lu *et al*. 2017; Delecroix *et al*. 2018, 2019; Jaspers *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a) collected data 30 minutes after sessions, two PQ studies (Fanchini *et al*. 2018; McCall *et al*. 2018b) collected data within 30 minutes after sessions, and only one PQ study (Malone *et al*. 2017) collected data immediately after sessions.

Injury data collection

All except one PQ study (Raya-González *et al*. 2018) consulted the football consensus statement by Fuller *et al*. (2006); that PQ study (Raya-González *et al*. 2018) consulted UEFA model for football injury studies by Hagglund *et al*. (2005) (Table 4). One PQ study (McCall *et al*. 2018a) consulted both models. Three PQ studies (Bowen *et al*. 2017, 2019; Delecroix *et al*. 2019) investigated both contact and non-contact injuries, whereas the remaining studies investigated non-contact injuries only. Despite consulting similar consensus statements, studies did not all report on the same injury data collection methods. Four PQ studies (Bowen *et al*. 2017, 2019; Malone *et al*. 2017; Jaspers *et al*. 2018) collected information on injury type, site, and mechanism; two PQ studies (Fanchini *et al*. 2018; McCall *et al*. 2018b) collected information on injury type, site, and circumstance (match/training). Only two PQ studies (Delecroix *et al*. 2018; Fanchini *et al*. 2018) specified how recurrent injuries were dealt with (recorded as new injuries).

Handling of missing data

Five PQ studies (Ehrmann *et al*. 2016; Bowen *et al*. 2017, 2019; Jaspers *et al*. 2018; McCall *et al*. 2018a) experienced missing training/match data and described data estimation methods (Table 5). Two such studies (Bowen *et al*. 2017, 2019) estimated missing training data using squad averages for drills completed, and individual season averages while considering game time for missing match data. The latter study (Bowen *et al*. 2019) further estimated data during international breaks using squad averages of other international players. One PQ study (Ehrmann *et al*. 2016) estimated missing match data by calculating positional averages and extrapolating according to playing time, quality of opposition, and phase of season/game for players who completed at least one full half in a preseason game. Another PQ study (Jaspers *et al*. 2018) estimated training data using averages of players with similar position in the same training session and used averages from friendly matches for match data estimation. The last PQ

Table 4. Injury classification.

Study	Injury consensus statement used?	Injury definition	Contact and/or non-contact	Injury information collected	Time loss classification	Diagnosis by:	RTP decision by:
Bowen *et al.*[36]	FIFA ICG consensus statement[48]	One that causes any absence from future football participation	Both	• Injury type (description) • Site (injury location) • Mechanism (contact vs non-contact)	• Minimal: 1–3 d • Mild: 4–7 d • Moderate: 1–4 w • Severe: >4 w	Academy doctor and senior chartered physiotherapist	-
Bowen *et al.*[37]	FIFA ICG consensus statement[48]	One that caused any absence from future football participation	Both	• Injury type (description) • Site (injury location) • Mechanism (contact vs non-contact)	• Minimal: 1–3 d • Mild: 4–7 d • Moderate: 1–4 w • Severe: >4 w	Club doctor and senior chartered physiotherapist	-
Delecroix *et al.*[38]	FIFA ICG consensus statement[48]	Any physical complaint sustained by a player that resulted from a football match or training, which made the player unable to participate in future football training or a match	Non-contact	-	-	Club medical doctor	-
Delecroix *et al.*[39]	FIFA ICG consensus statement[48]	Any physical complaint sustained by a player that resulted from a football match or training, which made the player unable to participate in future football training or a match	Both	-	-	-	-
Erhmann *et al.*[40]	FIFA ICG consensus statement[48]	Only non-contact soft tissue injuries that prohibited the injured player from participating in at least 1 match	Non-contact	-	-	-	-
Fanchini *et al.*[41]	FIFA ICG consensus statement[48]	When player is unable to take a full part in future football training or match due to physical complaints	Non-contact	• Circumstance (match or training) • Injury type • Injury location	-	Same club doctor	-
Jaspers *et al.*[42]	FIFA ICG consensus statement[48]	Time loss overuse injury (repeated micro-trauma without a single, identifiable event) that resulted in a player being unable to take a full part in training or match	-	• Location/Site • Type • Mechanism	-	-	-
Lu *et al.*[43]	FIFA ICG consensus statement[48]	Any physical complaint sustained from a match or training session resulting in time loss	Non-contact	-	-	-	-
Malone *et al.*[44]	FIFA ICG consensus statement[48]	All injuries that prevented a player from taking full part in all training and match-play activities typically planned for that day, and prevented participation for a period greater than 24 h	-	• Type • Site • Mechanism	• Low: training modification or 1–3 missed training sessions • Moderate: 1–2 w • High: 3 + w	-	-
McCall *et al.*[45]	FIFA ICG consensus statement[48]	Player unable to take full part in future on-field training or match due to physical complaints from a non-contact mechanism	Non-contact	• Circumstance (match or training) • Injury type • Injury location	• Slight: 1–3 d • Minor: 4–7 d • Moderate: 8–28 d • Major: >28 d	National team doctor	National team doctor
McCall *et al.*[46]	FIFA ICG consensus statement[48] + UEFA study model[5]	Player unable to participate in future football or training due to physical complaints	Non-contact	-	-	-	Club doctor
Raya-Gonzales *et al.*[47]	UEFA study model[5]	An injury that occurred during a scheduled training session or match that caused absence from the next training session or match	Non-contact	-	-		Medical staff

FIFA ICG = Federation Internationale de Football Association Injury Consensus Group; UEFA = Union of European Football Associations

Table 5. Statistical methodology.

			ACWR definition						Testing for association			Testing for predictive ability	
Study	Power analysis	Missing data estimation	1:2RA	1:3RA	1:4RA	Other	WL data pooling	Data testing	Measures of strength of association	Regression analysis	Effect size interpretation	Test(s) used	Benchmark for 'good' test
Bowen *et al.*[36]	-	• Training = squad averages for drills completed • Match = individual season averages (minimum three matches) while considering individual game time	-	-	✓	-	Low, moderate-low, moderate-high, high, and very high based on Z-scores	-	• RR (95% CI) • P-value	Binary logistic regression	• For an RR to be significant, 95% CI should not contain null RR of 1.00 • Significance at $P < 0.05$	N/A	N/A
Bowen *et al.*[37]	-	• Training = squad averages for drills completed • International = squad averages of other international players during international breaks • Match prior to 2015 = individual season averages (minimum three matches) while considering individual game time (if unavailable, based on friendly matches)	-	-	✓*	-	Chronic loads sorted into high and low by the median score for each variable ACWR sorted from very low to very high using Z-scores	-	• RR (95% CI) • P-value	Binary logistic regression	• For an RR to be significant, 95% CI should not contain null RR of 1.00 • Significance at $P < 0.05$	N/A	N/A
Delecroix *et al.*[38]	Power to detect a 1.2 RR = 0.50 Power to detect a 1.38 RR = 0.80	-	✓	✓	✓	Week: previous week	High, low, and reference according to one Z-score	-	• RR (95% CI) • P-value	N/A	-	• Sensitivity (95% CI) • Specificity (95% CI) • Positive and negative likelihood ratio • ROC, AUC, and Youden Index	-
Delecroix *et al.*[39]	-	-	✓	✓	✓	Week: previous week	-	• ACWR log-transformed when scores were not normally distributed	• RR (95% CI) • P-value	Poisson regression analysis	-	N/A	N/A
Erhmann *et al.*[40]	-	• Match = positional averages calculated and extrapolated according to playing time, quality of opposition, and phase of season/game for players who completed at least one full half in a preseason game	-	-	-	Injury block: average across 1 and 4 weeks leading up to injury Pre-injury block: average across 1 and 4 weeks before injury block Season average: average across beginning of season to point of injury	-	• Significant difference in GPS measure assessed with analysis of variance with repeated measure design	• Cohen's effect size • P-value	N/A	• Based on Cohen's rule of thumb: - 0.02 = small - 0.13 = medium - 0.26 = large • Significance at $P < 0.05$	N/A	N/A

(*Continued*)

Table 5. (Continued).

			ACWR definition						Testing for association			Testing for predictive ability	
Study	Power analysis	Missing data estimation	1:2RA	1:3RA	1:4RA	Other	WL data pooling	Data testing	Measures of strength of association	Regression analysis	Effect size interpretation	Test(s) used	Benchmark for 'good' test
Fanchini *et al.*[41]	-	-	✓	✓	✓	-	Based on percentiles: ● <15th = extremely low ● >15-50th = moderate low ● >50-85th = moderate high ● >85th = extremely high	● One-way ANOVA ● Pearson's moment of correlation ● VIF	● Magnitude-based inference ● Effect size (partial eta squared) ● OR (90% CI)	GEE with binary outcome distribution (injury: yes/no), logit link function	● Smallest beneficial and harmful effect for a risk ratio: <0.90 and >1.11 ● Thresholds for qualitative terms: - <0.5% = most unlikely - 0.5%–5% = very unlikely - 5%–25% = unlikely - 25%–75%, = possible - 75%–95% = likely - 95%–99.5% = very likely - >99.5% = most likely ● Significance at P < 0.05	● Sensitivity ● Specificity ● ROC and AUC (90% CI) ● Youden Index	AUC >0.70 and the lower CI >0.50
Jaspers *et al.*[42]	-	● Training = averages of players with similar position in the same training session ● Match = player's averages collected from friendly matches ● Total = 870 match + training sessions estimated	-	-	✓	-	Tertiles	● Data tested for normality and randomization	● Magnitude-based inference ● OR (90% CI)	GEE set for binary distribution (injury: yes/no), logit link function, first-order autoregressive working correlation structure	● Based on magnitude-based inferences: - <0.5% = most unlikely - 0.5%–5% = very unlikely - 5%–25% = unlikely - 25%–75%, = possible - 75%–95% = likely - 95%–99.5% = very likely - >99.5% = most likely ● Practical significance reached when true effect ≥75%	N/A
Lu *et al.*[43]	-	-	-	✓	-	-	-	● Repeated measures one-way ANOVA ● Post-hoc Bonferroni correction	● P-value	N/A	● Significance at P < 0.05	● Sensitivity (95% CI) ● Specificity (95% CI) ● Positive and negative likelihood ratios	-
Malone *et al.*[44]	Calculated statistical power = 90%	-	-	-	✓	-	4 groups, with the lowest workload range being the reference group	● Chi-squared analysis ● One-way ANOVA ● Correlation coefficients ● VIF	● OR (95% CI) ● P-value	Logistic regression	-	N/A	N/A

(*Continued*)

Table 5. (Continued).

Study	Power analysis	Missing data estimation	ACWR definition				WL data pooling	Data testing	Testing for association			Testing for predictive ability	
			1:2RA	1:3RA	1:4RA	Other			Measures of strength of association	Regression analysis	Effect size interpretation	Test(s) used	Benchmark for 'good' test
McCall *et al.*[45]	-	-	-	-	✓	-	-	-	• Cohen's effect sizes (90% CI)	-	• Based on Cohen's effect sizes - 0–≤0.2 = trivial - 0.21–≤0.6 = small - 0.6–≤1.2 = moderate - 1.21–≤2 = large - 2.1–≤4.0 = very large - ≥4.1 = nearly perfect	N/A	N/A
McCall *et al.*[46]	-	Based on player match or training values throughout season	✓	✓	✓	-	Based on percentiles: • <15th = extremely low • >15-50th = moderate low • >50-85th = moderate high • >85th = extremely high	• VIF	• Magnitude-based inference • RR (95% CI) • P-value	GEE with Poisson loglinear (link function) model with an autoregressive working correlations matrix	• Smallest beneficial and harmful effect for a risk ratio: <0.90 and >1.11 - <0.5% = most unlikely - 0.5%–5% = very unlikely - 5%–25% = unlikely - 25%–75%, = possible - 75%–95% = likely - 95%–99.5% = very likely - >99.5% = most likely • Significance at P < 0.05	•
Raya-Gonzales *et al.*[47]	Sensitivity • Specificity • Positive predictive values • ROC and AUC (95% CI) • Youden Index	AUC >0.70 and the lower CI >0.50	-	-	-	✓	-	-	-	• Magnitude-based inference • OR (90% CI)	GEE with binary outcome distribution (injury: yes/no), logit link function	• Smallest beneficial and harmful effect for a risk ratio = <0.90 and >1.11 - <0.5% = most unlikely - 0.5%–5% = very unlikely - 5%–25% = unlikely - 25%–75%, = possible - 75%–95% = likely - 95%–99.5% = very likely - >99.5% = most likely • Significance at P < 0.05	• ROC and AUC (90% CI)

RR = relative risk; ANOVA = analysis of variance; VIF = Variance Inflation Factor; ROC = Receiver operating characteristic curve; AUC = area under the curve; CI = confidence interval

study (McCall *et al*. 2018a) estimated missing values based on the player's match or training values throughout the season without further explanation.

Statistical analysis

Studies used similar statistical approaches (Table 5).

Time window

Ten PQ studies (Bowen *et al*. 2017, 2019; Malone *et al*. 2017; Delecroix *et al*. 2018, 2019; Fanchini *et al*. 2018; Jaspers *et al*. 2018; Wang *et al*. 2020) employed a 1-week absolute:4-week average rolling average ACWR. Four of these studies (Delecroix *et al*. 2018, 2019; Fanchini *et al*. 2018; McCall *et al*. 2018a) also explored 1:2 and 1:3 rolling average ACWR. One PQ study (Lu *et al*. 2017) only employed a 1:3 rolling average ACWR, and another PQ study (Ehrmann *et al*. 2016) also used a season average, where a ratio was created from averages made from the beginning of the season to the point of injury. Only one PQ study (Bowen *et al*. 2019) mentioned using uncoupled ACWRs.

Data testing

Seven PQ studies (Ehrmann *et al*. 2016; Lu *et al*. 2017; Malone *et al*. 2017; Fanchini *et al*. 2018; Jaspers *et al*. 2018; McCall *et al*. 2018a; Delecroix *et al*. 2019) described data testing. Two PQ studies (Jaspers *et al*. 2018; Delecroix *et al*. 2019) tested for normality. Four PQ studies (Ehrmann *et al*. 2016; Lu *et al*. 2017; Malone *et al*. 2017; Fanchini *et al*. 2018) performed one-way ANOVA testing, three PQ studies (Malone *et al*. 2017; Fanchini *et al*. 2018; McCall *et al*. 2018a) assessed for multicollinearity with variance inflation factors (VIF), and two PQ studies (Malone *et al*. 2017; Fanchini *et al*. 2018) utilised Pearson's moment of correlation.

Workload classifications

Seven PQ study (Bowen *et al*. 2017, 2019; Malone *et al*. 2017; Delecroix *et al*. 2018; Fanchini *et al*. 2018; Jaspers *et al*. 2018; McCall *et al*. 2018a) created workload classifications. Three PQ studies (Bowen *et al*. 2017, 2019; Delecroix *et al*. 2018) classified workload based on z-scores, two studies (Fanchini *et al*. 2018; McCall *et al*. 2018a) divided data into percentiles (<15th, 15–50th, 50–85th, >85th), one study (Jaspers *et al*. 2018) grouped data into tertiles, and another (Malone *et al*. 2017) into quartiles. Only one PQ study (Bowen *et al*. 2019) described classifying chronic workloads into high and low using median scores, in order to create distinctions between ACWRs with high and low chronic loads.

Analysis

Studies also used similar regression analysis techniques. Of the eight PQ studies (Bowen *et al*. 2017, 2019; Malone *et al*. 2017; Fanchini *et al*. 2018; Jaspers *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a; Delecroix *et al*. 2019) that performed regression analysis, four studies (Fanchini *et al*. 2018; Jaspers *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a) used generalised estimating equations (GEE), with the four remaining studies (Bowen *et al*. 2017, 2019; Malone *et al*. 2017; Delecroix *et al*. 2019) using binary logistic regression analysis. Two PQ studies (Ehrmann *et al*. 2016; McCall *et al*. 2018b) employed Cohen's effect sizes without regression analysis. Studies utilising GEE all used the disputed magnitude-based inferences (MBI; Batterham and Hopkins 2006) to interpret effect size (Hopkins 2006). Other studies favoured odds ratios and relative risk for effect size interpretation.

Five PQ studies (Lu *et al*. 2017; Delecroix *et al*. 2018; Fanchini *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a) tested the predictive ability of ACWR; four PQ studies (Delecroix *et al*. 2018; Fanchini *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a) utilised receiver operator characteristic curve (ROC) with area under the curve (AUC). Three (Delecroix *et al*. 2018; Fanchini *et al*. 2018; McCall *et al*. 2018a) of these also calculated the Youden index. Four PQ studies (Lu *et al*. 2017; Delecroix *et al*. 2018; Fanchini *et al*. 2018; McCall *et al*. 2018a) also calculated sensitivity and specificity.

Discussion

The aim of this systematic review was to evaluate current research practices of studies investigating the relationship between ACWR and injury incidence by identifying methodological variation between existing literature. Overall, there was notable heterogeneity in the methodological approaches between studies, including studies' ACWR definition, use of internal and external workload measures, statistical methods, injury definition, and handling of missing data. The qualities of included studies were all poor; frequently missed criteria included the representativeness of both exposed and non-exposed cohort, demonstration of lack of injury at the start of the study, reporting of de-identified handover of injury data for assessment of outcome, and comparability. Studies also had various similarities in their methodological approaches. The majority of studies used GPS technology to quantify external workload and based investigation on different running intensities. Studies also gravitated towards RPE to quantify internal workload. Further, studies shared similarities in statistical tools – rolling averages (particularly of a 1:4-week ratio), workload classification/grouping, GEE, and MBI being popular tools.

ACWR definition

There were consistencies in the way studies calculated ACWR, with all studies utilising the rolling average method. Though past studies from elite cricket and rugby have found rolling average ACWRs to be useful for injury prevention (Hulin *et al*. 2016; Lolli *et al*. 2018), recent studies have suggested that this traditional method of calculating ACWR 'mathematically couples' the acute and chronic portions of the ACWR (Lolli *et al*. 2017, 2018; Windt and Gabbett 2018) inducing spurious correlations within the calculations. To overcome this, 'uncoupled' ACWR assessment, such as that employed by Bowen *et al*. (2019), can be applied. Unfortunately, extensive literature highlight pitfalls inherent to the use of ratios that may not be resolved through use of uncoupling (Curran-Everett 2013).

Future studies may also explore different acute:chronic time windows. As stated by Carey *et al*. (2016), the choice of acute and chronic timeframes used in conventional ACWRs lack rationale and are based on studies on swimmers (Calvert *et al*. 1976). As such, the effects of ACWRs calculated based on a daily basis are not well understood, especially in football. The study by Carey *et al*. (2016) has suggested that an ACWR using a 3-day-to-3-week ratio best explains injury likelihood in Australian football, thereby reinforcing the relevance of future football ACWR studies to also explore alternative ACWRs. Studies should also be aware that training scheduling may confound relationships between ACWR and injury (Bornn *et al*. 2019). As such, future studies may look to incorporate training or match scheduling in ACWR time window definitions. In addition, the fluctuation in training intensity to fit around match play schedules in elite football should be acknowledged as a natural limitation to ACWR calculations.

On the other hand, a recent study by Impellizzeri *et al*. (2020) found that acute to random chronic workload ratios were as associated with injury as actual ACWRs. This paper highlights a need for further research regarding the complex aetiological mechanisms behind workload and injury risk. Hopefully in the future, other forms of ACWR definition based on a more rigorous and thoroughly understood aetiological framework may be applied and investigated.

Recent studies have also suggested use of a non-linear method of representing ACWR, as opposed to the linear 'rolling average' method employed by the included studies. This non-linear method is the exponentially weighted moving averages (EWMA) method. EMWA has been suggested to be more sensitive (Murray *et al*. 2016) and better suited (Menaspa 2017) to identifying increases in injury risk with higher ACWR, as it considers the time frame in which a stimulus occurs and the decaying nature of fitness and fatigue over time. This is achieved by assigning a decreased weighting for each older load (Williams *et al*. 2016). In practice, research in Australian-rules football comparing rolling average and EWMA ACWRs has suggested that the latter method is more sensitive to injury risk increases (Menaspa 2017). As such, future studies may consider comparing EWMA to the conventional rolling average approach in the context of elite male football.

Internal workload measure

RPE was the most investigated internal workload measure (nine PQ studies (Lu *et al*. 2017; Malone *et al*. 2017; Delecroix *et al*. 2018, 2019; Fanchini *et al*. 2018; Jaspers *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a, 2018b)) in the included studies. RPE is a standard tool of the American College of Sports Medicine to assess fatigue, commonly assessed with the Borg scale (Utter *et al*. 1997). It has been previously evaluated as a valid and reliable in football to quantify global internal training load (Impellizzeri *et al*. 2004; Haddad *et al*. 2017) with positive correlations with physiological measures such as heart rate, blood lactate, VO2 max, ventilation, and respiratory rate (Chen *et al*. 2002). In addition, studies have also suggested that subjective measures may also be more sensitive and consistent than objective measures (Saw *et al*. 2015) especially as they may be better at accounting for the allostatic stress athlete experience in mixed training sessions (e.g. tactical, strength, fitness; Delaney *et al*. 2018; Graham *et al*. 2018) and mental load (e.g. learning new skills, competing against unfamiliar opponents; Weston *et al*. 2015; Barrett *et al*. 2018).

Despite advantages of RPE-based variables and its ubiquity in sports research, the authors of this review are not aware of any papers that highlight the mechanisms underpinning the rationale behind why RPE should be related to injuries. Theoretical models relating workloads and injury highlight a continuously fluctuating injury risk (Gabbett *et al*. 2016), thus illustrating the importance of collecting data frequently enough to observe changes in these variables (Windt e. 2018). Whilst RPE-based variables are valuable candidates for further investigation, future studies can also consider other subjective (e.g. perceived recovery or holistic wellness-based) and quantifiable (e.g. heart rate-based) internal workload measurements that may either better reflect tissue-specific or stress-strain characteristics of football activity, or capture not only the association of training stimulus on an athlete's injury risk, but also the effects of sleep, nutrition, and psychology.

External workload measures

Of the studies that investigated external workload, six studies (Ehrmann *et al*. 2016; Bowen *et al*. 2017, 2019; Lu *et al*. 2017; Jaspers *et al*. 2018) used GPS-derived variables. Reliability of GPS data is influenced by movement type, velocity, duration, and sampling rate (Aughey 2011). These disadvantages apply to football-specific tasks (with frequent changes of direction) and training (repetitive tasks performed in a small field area) (Portas *et al*. 2010; Jennings *et al*. 2010, 2010; Duffield *et al*. 2010). In fact, Ironside *et al*. (2017) reported location error of GPS data even under ideal conditions (little obstruction from sky-view) can be up to 13.38 m with stationary GPS collars. Additionally, GPS data lacks quantification of other loading activities (e.g. jumping, ball interaction, player interaction) (Aughey 2011). Studies should be cautious of their GPS calibration, especially due to the variation in sampling rates observed in this review (5 Hz, 10 Hz, and 15 Hz). Research has suggested that 10 Hz devices are more reliable and valid than 5 Hz devices, but an increase to 15 Hz may have no additional benefit (Scott et al. 2016). Use of accelerometer data may also be considered to quantify load, such as in studies by Bowen *et al*. (2017), Bowen *et al*. (2019), Ehrmann *et al*. (2016), and Lu *et al*. (2017). In addition, future studies may also consider using camera systems to track external workload, as done in Bowen *et al*. (2019). However, studies should be aware that the appropriate interchangeability between camera systems and GPS data depends on the calibration equations used, with certain equations yielding a typical error of the estimate to be of moderate magnitude (Buchheit *et al*. 2014). As such, calibration methods of video and GPS data should be reported to prevent bias in results. Football is a physically demanding sport with intermittent exercise and periods of maximal or near-maximal intensity activity interspersed throughout. Using an accelerometer may supplement data from conventional GPS units, as

accelerometer data is collected at higher sampling rates and are movement-based (as opposed to position-based) (Alexander *et al*. 2016; Nicolella *et al*. 2018).

Differences in speed thresholds applied to GPS data contribute to heterogeneity of study outcomes, making combined interpretation of results difficult. Studies only investigating one running intensity lack sensitivity to the effects of running intensities on fatigue; studies with multiple speed zones should ensure that appropriate thresholds are chosen. As such, individualized speed thresholds can be established as demonstrated by a case-cohort example (Lovell and Abt 2013). Although this method has practical challenges, it eliminates the variation that exists even in studies of football-specific running profiles (Rampinini *et al*. 2007; Bradley *et al*. 2009; Miñano-Espin *et al*. 2017).

Statistical methods

Future studies should also employ appropriate statistical tools with rigorous defence of choices. In particular, it is necessary to ensure that the data meet specific criteria (or assumptions) before using certain statistical tests or regression analysis. Longitudinal data analysis can be most effective when the statistical methods are aligned to the theoretical mechanisms of workload and injury. In the included studies, only three studies (Malone *et al*. 2017; Fanchini *et al*. 2018; McCall *et al*. 2018a) tested for multicollinearity, and only one study (Jaspers *et al*. 2018) tested for randomisations. Due to the multifactorial and dynamic complexity of injury risk, and the intra- and inter-athlete fluctuations in injury risk, future studies may consider testing for independence of data. Data that does not meet the assumption of normality can also give poor results for certain types of test. Independent samples t-test, hierarchical linear modelling, ANCOVA, goodness of fit test, and linear modelling in the Gaussian framework rely on normality – a distribution that may not apply in relation to workload.

Varying methods of discretization were also employed in the included studies, including use of medians, percentiles, and z-scores to sort ACWRs into categories from low to high. Bowen *et al*. (2019) adopted a method of discretization where the chronic workloads of the ACWRs were also discretized. This more nuanced approach facilitates investigation of how an athlete's chronic workload (fitness) superimposed on spikes or troughs in workload (fatigue) can affect injury risk. In other words, this approach may distinguish between the effects of increasing workload (ACWR > 1) in athletes who have already endured a long period of high training stimulus (high chronic workload) and the effects of increasing workload in athletes who have had low chronic workloads (after international break, for example). Future studies should also be aware of criticisms against discretization of a continuous risk factor. Carey *et al*. (2018) highlight that discretization of cumulative loads into low, medium, and high categories by percentiles, for example, assumes that athletes in each category have equal risk regardless of how broad the percentile cut-offs for that category was set to be. This loss of information may lower statistical power, increase likelihood of false negatives, and inflate discover rates (Altman and Royston, 2006; Bennette and Vickers 2012). Instead of discretization, use of continuous methods were suggested by Carey *et al*. (2018)

In addition, the MBI is an inference method for measuring the probability that an effect is beneficial, harmful, or trivial, and is widely used in the sport science community (Hulin *et al*. 2016; Hopkins 2006). Though it is practical, there is a strong and ongoing debate on whether this methodology is statistically accurate (Johnson and Wichern 2002; Borg *et al*. 2018; Sainani *et al*. 2019).

Injury definition

Injury definition will inevitably influence findings comparing ACWR and injury risk. All included studies used the injury definitions and data collection consensus statement by Fuller *et al*. (2006) and adopted a time loss classification approach to stratifying injury severity or duration. Following a standardised approach allows better comparisons between studies (Hopkins and Batterham 2018). However, studies still differed in the type of injury collected for analysis; only three studies (Bowen *et al*. 2017, 2019; Delecroix *et al*. 2019) included both contact and non-contact injuries. Seven studies (Ehrmann *et al*. 2016; Lu *et al*. 2017; Delecroix *et al*. 2018; Fanchini *et al*. 2018; Raya-González *et al*. 2018; McCall *et al*. 2018a, 2018b) only collected and analysed non-contact injuries as this injury mechanism accounted for the majority of football injuries and are largely preventable (as opposed to contact injuries) (Finch 1997). It is only speculative without established aetiological mechanisms associating workload and injury to determine whether this choice may lead to potentially inaccurate estimations of the effects of ACWR on overall injury.

Handling of missing data

Lastly, included studies differed over their handling of missing data. Only five of the included studies (Ehrmann *et al*. 2016; Bowen *et al*. 2017; Malone *et al*. 2017; McCall *et al*. 2018a, 2018b) specifically mentioned collecting data for gym sessions and rehabilitation/recovery sessions, and four studies (Bowen *et al*. 2017; Lu *et al*. 2017; McCall *et al*. 2018b; Delecroix *et al*. 2019) did not specify whether data collected from matches were friendly or competitive matches. Although collecting and comparing data from different exposure types (such as on-pitch versus weights session in the gym) presents other practical challenges, data collected without consideration for gym/recovery sessions may not represent the actual workload a player is subjected to.

The five studies (Ehrmann *et al*. 2016; Bowen *et al*. 2017, 2019; Jaspers *et al*. 2018; McCall *et al*. 2018a) that described data estimation had different approaches to this process – including using squad averages (Bowen *et al*. 2017, 2019), positional averages (Ehrmann *et al*. 2016; Jaspers *et al*. 2018), individual averages from friendly matches (Jaspers *et al*. 2018), and individual seasonal averages (Bowen *et al*. 2017, 2019; McCall *et al*. 2018a), with or without consideration of factors such as playing time, opposition quality, and phase of game/season (Ehrmann *et al*. 2016). Bowen *et al*. (2019) performed data estimation on one full season of missing game data based on individual season game averages from

the two subsequent seasons. As the missing data is not random, biases caused by the missing data can only be addressed through sensitivity analysis examining the effects of data imputation on results (Sterne *et al*. 2009).

If appropriate, data estimation may consider taking into account the fact that a player's activity profile depends on their role in their team. Further, Squad averages do not account for different activity profiles between positions (Bradley *et al*. 2010; Clemente *et al*. 2013; Baptista *et al*. 2018). Positional averages may overcome this, but individual averages may further reduce inter-positional variations. Individual averages from friendly fixture data may cause underestimation – players may not be as competitive, and key players may be rested more in preparation for important matches. Studies have also suggested that activity profiles of professional football players change significantly with the activity profiles and quality of opponents (Rampinini *et al*. 2007) Therefore, future studies should aim to use individual averages and incorporate a multitude of activity-modifying factors into data estimation.

In close relation to missing data and data estimation, future studies may also consider reporting on data sparseness, and whether any testing or adjusting were done to reduce sparse data bias. For example, Greenland *et al*. (2016) suggested use of penalisation as a diagnostic tool for as well as a solution for potential sparse data bias.

Limitations

Limitations of this systematic review must be acknowledged. Only English language articles were considered, and only studies investigating elite male football players were included. In addition, limitations were found in identifying risk of bias using the NOS – some elements of the NOS were not completely applicable with the nature of the included studies. Hence, these NOS categories were interpreted and adapted to be better suited to the included papers. The included studies were still examined based on methodological limitations or concerns.

Conclusion

The systematic review gathered 12 studies, all of poor study quality, attempting to assess the relationship between ACWR measures and injury in elite male football with methodological heterogeneity in multiple study features. These features included differences in ACWR time windows, GPS-derived measures, GPS equipment setup and running speed/intensity thresholds, exposure types measured, injury types investigated, data estimation methods, and statistical analysis. These findings fall in line with the growing evidence demonstrating the limitations of ACWR as a metric for determining injury risk (Australian Institute of Sport. Athlete Management System. Recent Developments in Athlete Workload and Health Monitoring 2020; Wang *et al*. 2020; Impellizzeri *et al*. 2020). In order for a meaningful discussion on the implications of results gathered, future studies may consider several options to further bolster methodological rigor. First, they may consider consulting existing international guidelines for developing prognostic studies to improve quality of reporting. From this basis, further research may strive to elucidate aetiological mechanisms that underpin the relationship between workload and injury risk. An improved understanding of underlying theoretical framework may then enhance decisions downstream, such as the mathematical definitions of ACWR, the type of internal or external workload measures used, and the statistical methods applied.

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Study topic development and search strategy – AW, NH, JH, KOK
Search – AW, NH, JH
Third reviewer – KOK
Data analysis – AW, NH, JH, KOK, GB
Write up – AW, KOK
Review/Feedback – AW, NH, JH, KOK, GB

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